

Course content and style

International Marketing Analytics



Dr Arne Floh, Senior Lecturer in Marketing
WU Vienna, Institute for International Business
arne.floh@wu.ac.at
[Office hours: book online \(Google Calendar\)](#)

WINTER TERM 2026/27



The journey begins



Module Convenor - Dr Arne Floh



Personal Information

- **Senior Lecturer in International Marketing - WU Vienna**
- Deputy Program Director Export- und Internationalisierungsmanagement
- WWW:
<https://www.wu.ac.at/en/welthandel/arne-floh/>
- LinkedIn: <http://www.linkedin.com/arnefloh>

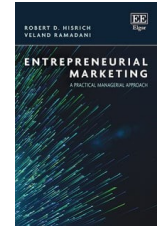
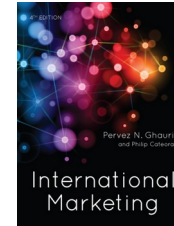
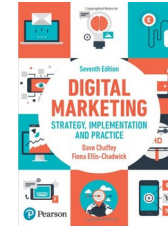
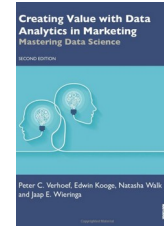
Contact Information

- T: 01 31336 6367
- E: arne.floh@wu.ac.at
- Room D1.5.068
- Office Hours: by appointment

Module Convenor - Dr Arne Floh



- Work Experience
 - International & Industry Background
 - Teaching Experience > 25 Years
- Teaching
 - International Marketing
 - International Business Projects
 - Marketing Research and Marketing Analytics
 - Global Branding
 - Supervision (BSc, MSc, PhD)
- Research
 - Dynamic Capabilities and Firm Performance
 - GenAI and Marketing
 - SMEs and Internationalization
- Private
 - Football
 - Electronic Gadgets
 - Austrian Wine





Marketing analytics turns marketing data into decisions on markets, prices, channels and budgets, and checks whether they worked.



One budget question: how should a fixed media budget be split across countries and channels?



Marketing mix modelling is back at the centre since privacy rules weakened user-level tracking. Managers who can read and challenge it own the budget debate.



One running case: Alpenglow, a Vienna-based premium chocolate brand selling in Austria, Germany, France, Italy, the Netherlands and Poland.



Python with AI assistants: the assistant does most of the typing; you specify, run, check and communicate the analysis.

Five sessions, five decisions

Session 1	Session 2	Session 3	Session 4	Session 5
Tue 6 Oct Is our price too high in Poland? → Price and promotion elasticities	Tue 13 Oct What did TV do for us in Germany? → Adstock, saturation and ROAS	Tue 20 Oct Where should the next euro go? → Budget allocation across countries and channels	Tue 27 Oct Can we trust the model? → Forecasts, geo-lift tests and attribution	Tue 3 Nov Board meeting → Groups pitch and defend their 2027 budget

The question behind all five sessions: how should Alpenglow split a fixed media budget across six countries and five channels?

On successful completion of this module, students will be able to:



Model the market

- 1 Explain marketing mix modelling:** carryover, saturation, baseline versus incremental sales, ROAS versus marginal ROAS.
- 2 Estimate response models** in Python, from OLS and log-log elasticities to fixed effects and Bayesian MMM.



Decide on budgets

- 3 Allocate budgets** across countries and channels and defend the allocation under uncertainty.
- 4 Forecast sales** and evaluate the forecasts as the baseline for planning.



Test the evidence

- 5 Design experiments:** analyse a geo-lift test and use it to validate and calibrate an MMM.
- 6 Compare attribution:** explain why last-touch, regression-based and Shapley attribution disagree across markets.

< > Work with tools

7 Work reproducibly: use Python, Positron, Quarto and GitHub to build analyses that colleagues can rerun and audit.

8 Use AI assistants responsibly: direct an AI coding assistant, verify its output and document the prompts behind an analysis.

👤 Work with people

9 Communicate: present analytical findings to a board audience in a written report and a pitch.

10 Work in teams: collaborate in version-controlled projects and take collective, data-driven decisions.

11 Think critically: evaluate data sources, model assumptions and findings in an international marketing context.

Every session

Concept lecture
60 min

Guided lab
90 min

Team exercise
60 min

**Debrief
and quiz**

Between sessions



- Read the book chapters and materials in advance
- Online quiz after Sessions 1 to 4
- Lab check-ins after Sessions 2 and 4
- Group project in your team repository
- Q&A forum on Canvas; consulting hours on request



In the classroom

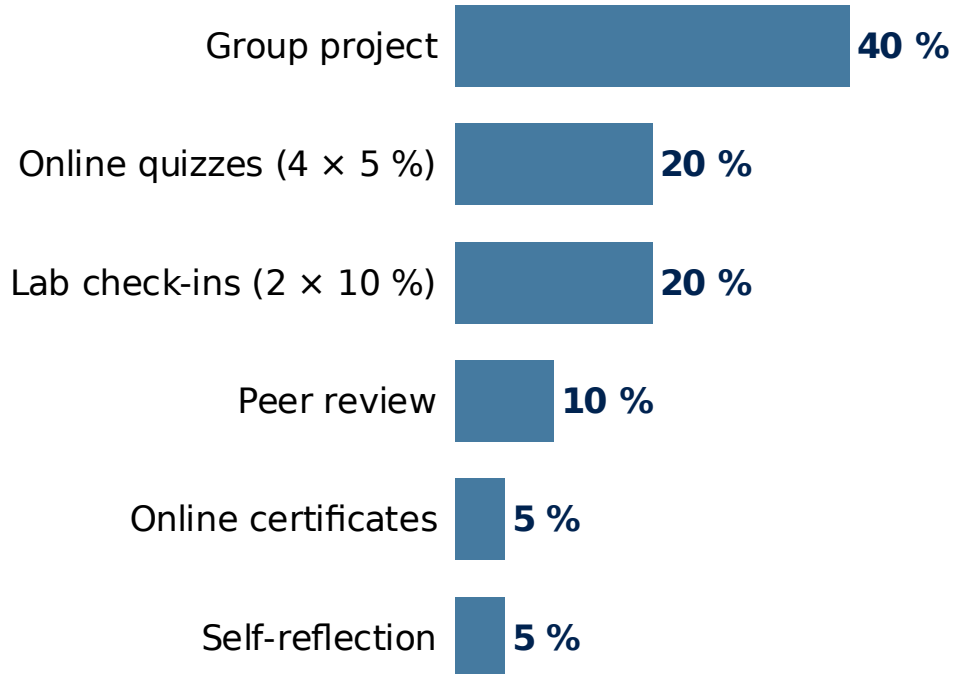


- Five Tuesdays, 6 October to 3 November 2026
- Bring your laptop with Python, Positron, Quarto and GitHub
- AI-assisted coding: you direct, the assistant types, you check
- Pull, commit and push from Session 1

Module roadmap

Session	Date	Topic	Methods	Chapter
1	Tue 6 Oct	Course content and style, toolkit, regression foundations	Descriptive statistics, OLS, log-log elasticities, dummies and interactions	1, 11
2	Tue 13 Oct	Marketing mix modelling I: one country	Adstock, saturation, decomposition, ROAS, diagnostics	3
3	Tue 20 Oct	Marketing mix modelling II: many countries, one budget	Panel fixed effects, hierarchical Bayesian MMM, budget optimisation	3
4	Tue 27 Oct	Forecasting, experiments and attribution	Time series regression, difference-in-differences, logistic regression, Shapley values	9, 4
5	Tue 3 Nov	Group project pitches and wrap-up	Board presentation, ground-truth reveal	11

Weight in the final grade



Rules



- **Groups:** 5 students; enrol on Canvas by the end of Session 1
- **Grades:** WU scale, 1 from 90 %, 2 from 80 %, 3 from 70 %, 4 from 60 %
- **Attendance:** at least 80 % of sessions
- **Late work:** minus 10 points per day unless agreed
- **AI tools:** expected; you own every number you report

5 h

Kaggle Learn: Python. Free course with a certificate on completion.

1 h

GitHub Skills: Introduction to GitHub. Free, runs in the browser.

45 min

Course environment. Install Python (via uv), Positron, Quarto and GitHub Desktop with the setup guide, then run the installation check.



Deadline: upload both certificates and a screenshot of the passing installation check on Canvas by Mon 12 Oct, 11:59 pm.

Quizzes and lab check-ins

Online quizzes



- After each of Sessions 1 to 4
- Open Wednesday 9 am, close Monday 11:59 pm
- 10 multiple-choice questions in 10 minutes
- Closed book: concepts, not code

5 % per quiz, 20 % in total

Lab check-ins



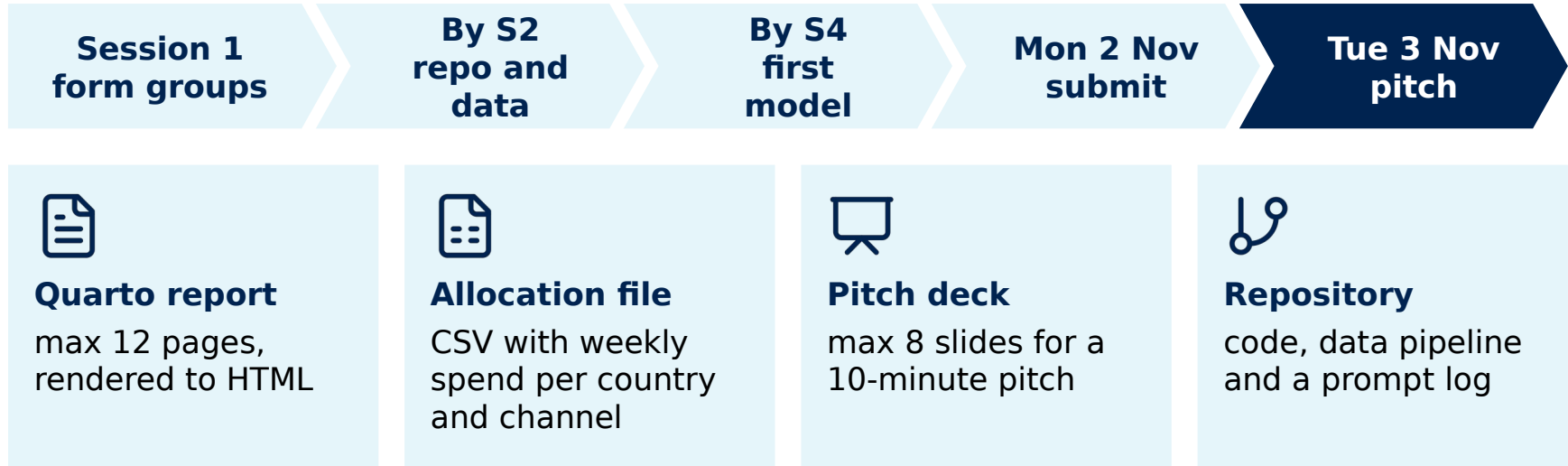
- After Sessions 2 and 4
- Replicate the lab for another country or dataset
- Submit notebook, HTML, result files and prompt log
- Auto-checked for plausibility, reviewed for interpretation

10 % per check-in, 20 % in total

Both are individual: quizzes check that you understand the concepts, check-ins that you can run, check and explain an analysis on your own.

Group project

Timeline



The task: as Alpenglow's analytics team, recommend to the board how the 2027 media budget should be split across six countries and five channels.

- **Who:** every group member rates the other members on a 0 to 10 scale
- **Criteria:** tasks carried out, deadlines met, quality of work, communication and respect, attendance, pre-agreed rules
- **Confidential:** feeds the peer review component (10 %)
- **When:** submit on Canvas with the final project

Peer-Rating-Form

Each group member is required to submit a peer rating form. This form must contain an evaluation (i.e. a grade) of each group member. Do not assign yourself a grade as this would defeat the idea of peer rating. **The form will be kept in strict confidence.**

In the spaces provided below, please fill in your name, choose your group, and write complete names of all of your group members. Finally, assign a grade on a scale from 0 to 10 to each of your group members (**10 being the highest** and 0 being the lowest grade). Please assign grades based on the following criteria:

- → Willingness to carry out assigned tasks
- → Ability to meet deadlines
- → Quality of the individual's work
- → Communication ability and respectful treatment of others
- → Availability for and attendance at group meetings
- → Following the pre-agreed rules

Your Name: → Please enter your name

Your Group: → Choose your group

Group Member Names	Grade
1.Group Member Name	Insert grade
2.Group Member Name	Insert grade
3.Group Member Name	Insert grade
4.Group Member Name	Insert grade
5.Group Member Name	Insert grade



VIENNA UNIVERSITY OF
ECONOMICS AND BUSINESS

**INSTITUTE FOR INTERNATIONAL
BUSINESS**

Welthandelsplatz 1, 1020 Vienna,
Austria

DR ARNE FLOH

Senior Lecturer in Marketing

F +43-31336-6367

arne.floh@wu.ac.at

www.wu.ac.at/en/welthandel/arne-floh

[Office hours: book online](#)